

Bitwise linear trail through **QALQAN** found by SAT

CiVerLy

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1 QALQAN

Maximal linear correlation: 2^{-48} [illegible]

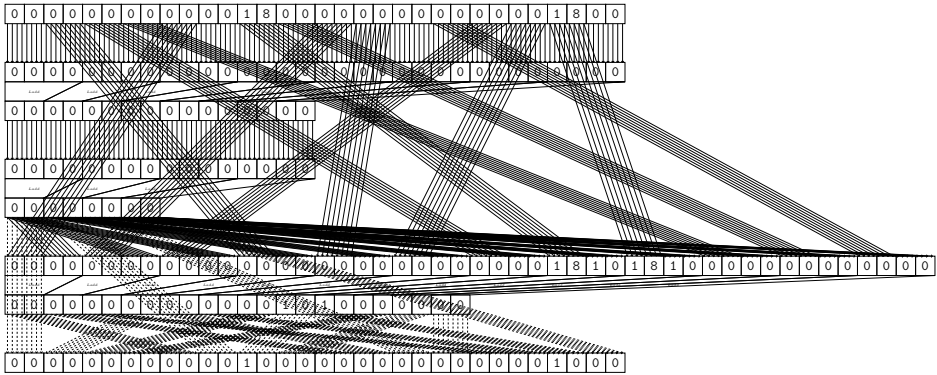
2 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

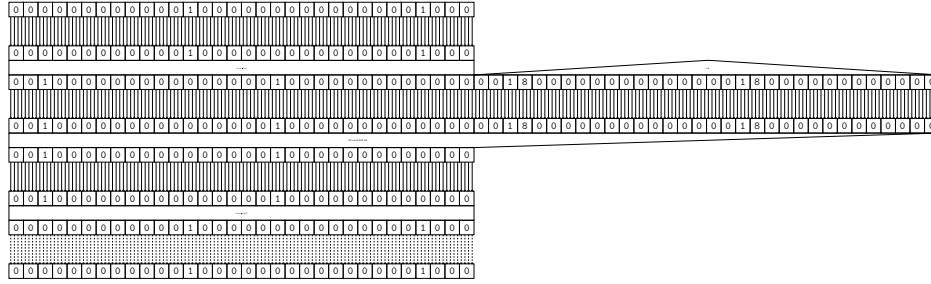
0	0	0	0	0	0	0	0	0	0	0	0	e	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	0	0
0	0	0	0	0	0	0	0	0	0	0	0	e	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0

3 L

Maximal linear correlation: $2^{-2.0}$



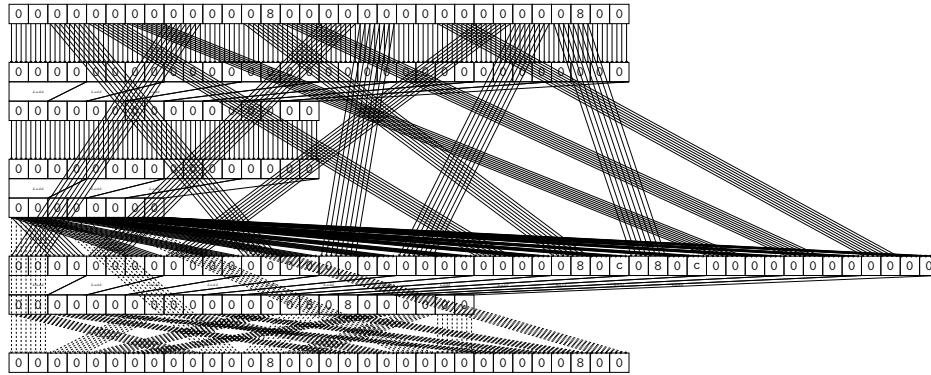
4 Add128

Maximal linear correlation: $2^{-2.0}$ 

5 SBoxLayer

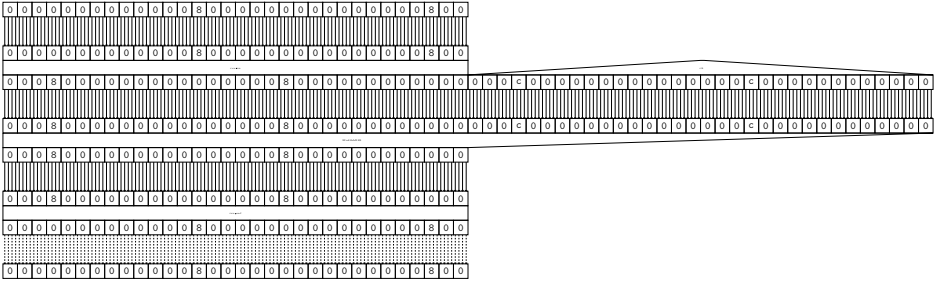
Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0

6 LMaximal linear correlation: $2^{-2.0}$ 

7 Add128

Maximal linear correlation: $2^{-2.0}$



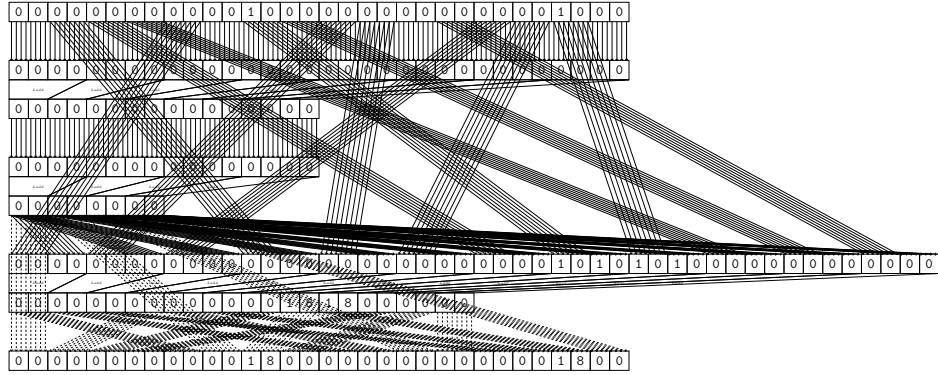
8 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

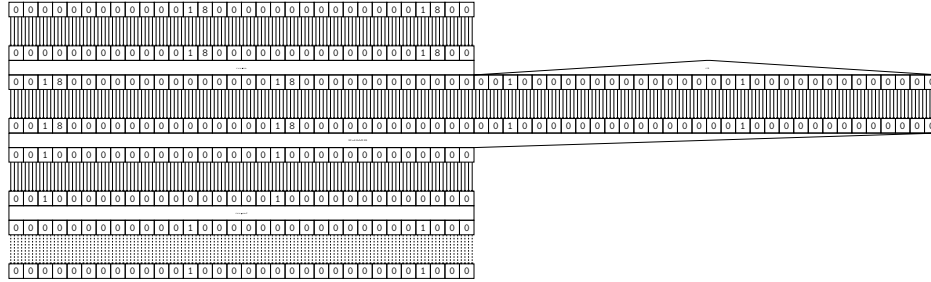
0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

9 L

Maximal linear correlation: $2^{-2.0}$



10 Add128

Maximal linear correlation: $2^{-2.0}$ 

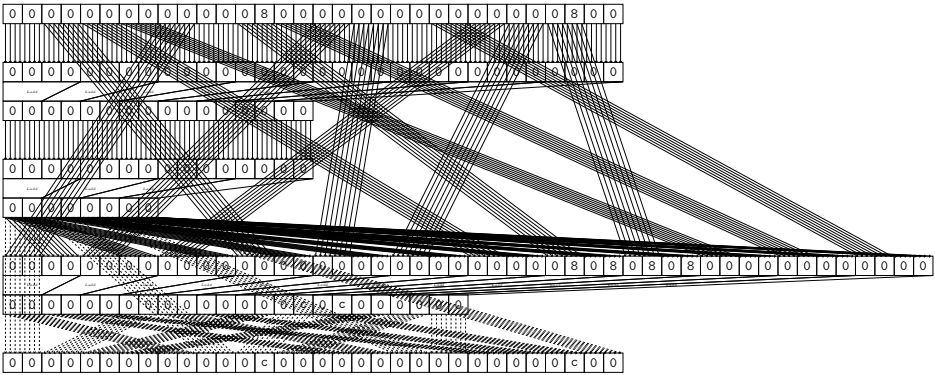
11 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0

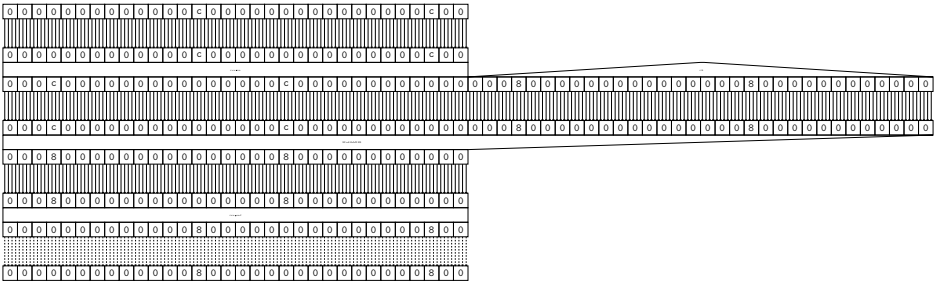
12 L

Maximal linear correlation: $2^{-2.0}$



13 Add128

Maximal linear correlation: $2^{-2.0}$



14 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0

15 L

Maximal linear correlation: $2^{-2.0}$ 